

SoulScope Evidence Marker Registry v0.1

Status: Canonical scientific implementation companion — final for adoption

Governing authority: SoulScope Canon v1.3

Upstream authority: Acoustic Parameter Registry v0.1

Downstream authority: Constellation Dimension Registry v0.1

Date: August 14, 2026

1. Purpose

The Evidence Marker Registry defines the intermediate observational layer between acoustic measurement and semantic reasoning.

An Evidence Marker answers:

What changed or remained consistent in the qualified acoustic evidence?

It does **not** answer:

What emotion is this person feeling?

What does this reveal about their personality?

Why did this happen?

The Evidence layer exists so that no raw acoustic feature maps directly to a Constellation meaning.

Canonical sequence:

Acoustic Measurement → Qualification → Evidence Marker → Evidence Ledger → Dimension Reasoning

2. Evidence Marker object

Every Evidence Marker SHALL produce:

```
evidence_id
marker_id
marker_version

scan_id
prompt_scope
```

```

time_scope
reference_scope

source_measurement_ids[]
source_feature_families[]

value
direction
magnitude
uncertainty

quality
coverage
agreement
baseline_trust

supporting_components[]
contradicting_components[]
missing_components[]
confound_flags[]

rule_id
rule_version

status
resolution_reason

```

Canonical statuses:

```

RESOLVED
PARTIALLY_RESOLVED
UNRESOLVED
SUPPRESSED
INVALID

```

3. Reference-aware effect representation

For a compatible feature f , prompt/segment a , and reference r :

$$d_{f,a|r} = \frac{\tilde{x}_{f,a} - \tilde{x}_{f,r}}{\max(1.4826 MAD_{f,r}, \epsilon_f)}$$

subject to versioned clipping and validity rules.

The reference may be:

- within-prompt;
- P1 within-session;
- established personal history;
- provisional personal history;
- validated compatible population reference.

The reference identity travels with the Evidence Marker.

No valid spread:

⇒ UNRESOLVED

not zero.

4. Composite Evidence

When an Evidence Marker uses multiple measurements, the first implementation may use a reliability-weighted robust aggregation:

$$E_m = \text{weightedMedian}(o_i d_i; w_i q_i)$$

where:

- d_i = standardized measurement effect;
- o_i = registered orientation for that marker;
- w_i = versioned engineering prior;
- q_i = measurement quality.

All w_i values remain engineering priors until calibrated on SoulScope data.

No weight becomes scientific truth merely because it is coded.

5. Evidence confidence

Evidence confidence SHALL consider independently:

$$C_E = f(Q, Cov, A, B, Conf)$$

where:

- Q = measurement quality;
- Cov = required-component coverage;

- A = agreement among eligible measurements;
- B = baseline/reference trust;
- $Conf$ = active confound burden.

A strong effect with poor quality may therefore remain unresolved.

6. Prosodic Evidence Family

EV_PRO_001 – PITCH_LEVEL_SHIFT

Definition: Qualified change in central F0 level relative to an explicit reference.

Inputs:

- F0 median;
- F0 mean where authorized;
- semitone-normalized F0.

Output:

direction = LOWER | HIGHER | NONE | UNRESOLVED

May support: Activation, Mobilization, contextual reconfiguration.

Cannot independently support: emotion, anxiety, confidence, deception, Processing Demand.

EV_PRO_002 – PITCH_RANGE_SHIFT

Definition: Change in robust F0 range.

Preferred form:

$$R_{F0} = P_{95}(F0) - P_5(F0)$$

preferably represented in semitones.

Output:

EXPANDED | COMPRESSED | SIMILAR | UNRESOLVED

May support: Range, expressive deployment, Mobilization, Flexibility.

EV_PRO_003 – PITCH_VARIABILITY_SHIFT

Definition: Change in robust within-condition F0 variability.

Candidate measures:

- MAD;
- SD where appropriate;
- calibrated contour variability.

May support: Stability, Range, modulation breadth.

EV_PRO_004 – PITCH_CONTOUR_DRIFT

Definition: Systematic F0 movement over the course of a prompt.

Requires: sufficient qualified temporal coverage.

May support: within-prompt drift, short-interval Sustainability.

Hard boundary: drift does not itself mean improvement, deterioration, escalation, or recovery.

7. Energy Evidence Family

EV_ENG_001 – LOUDNESS_LEVEL_SHIFT

Qualified change in central loudness/energy level.

May support: Mobilization, Activation.

Confounds include:

- microphone gain;
 - mouth-to-microphone distance;
 - automatic gain control;
 - room acoustics;
 - intentional speaking style.
-

EV_ENG_002 – LOUDNESS_RANGE_SHIFT

Change in robust loudness modulation range.

May support: Range, expressive deployment, contextual reconfiguration.

EV_ENG_003 – LOUDNESS_DRIFT

Systematic movement in qualified loudness over the prompt.

May support: short-interval maintenance and temporal-dynamics evidence.

8. Timing Evidence Family

EV_TIM_001 – RESPONSE_ONSET_SHIFT

Change in latency from capture-ready event to first qualified speech.

May support: temporal organization only.

Cannot independently mean:

- hesitation;
 - avoidance;
 - fear;
 - uncertainty;
 - deception;
 - Processing Demand.
-

EV_TIM_002 – PAUSE_LOAD_SHIFT

$$PL = \frac{\text{qualified silent-pause time}}{\text{qualified response interval}}$$

Output:

INCREASED | DECREASED | SIMILAR | UNRESOLVED

May support: Focus Continuity, Organization, Processing Demand candidates, Effort Cost candidates.

EV_TIM_003 – PAUSE_DURATION_SHIFT

Change in pause-duration distribution.

Candidate summaries:

- median;
 - P90;
 - robust dispersion.
-

EV_TIM_004 – PAUSE_DENSITY_SHIFT

Change in number of qualified pauses per unit of speech/responding time.

EV_TIM_005 – SPEECH_RATE_SHIFT

Change in versioned speech-rate estimate.

The unit and language dependency SHALL be recorded.

EV_TIM_006 – ARTICULATION_RATE_SHIFT

Change in articulation rate excluding qualified silent pauses.

Speech rate and articulation rate remain separate variables.

EV_TIM_007 – SPEECH_RUN_STRUCTURE

Describes the distribution of continuous speech-run durations.

Possible output:

```
LONGER_RUNS  
SHORTER_RUNS  
MORE_VARIABLE  
LESS_VARIABLE  
SIMILAR  
UNRESOLVED
```

EV_TIM_008 – OUTPUT_CONTINUITY

Composite descriptive Evidence representing sustained continuity of qualified speech production.

Candidate inputs:

- voiced/speech ratio;
- pause load;
- speech-run duration;
- pause density.

Requires at least two independently eligible timing measures.

EV_TIM_009 – TEMPORAL_FRAGMENTATION

Composite Evidence representing increased interruption or segmentation in the observable timing pattern.

This is not mental fragmentation.

It is not thought disorder.

It is not impaired attention.

9. Phonatory Evidence Family

EV_PHO_001 – CEPSTRAL_SHIFT

Qualified change in CPP/CPPS where the adopted implementation supports connected speech.

May support: phonatory-change Evidence and calibrated higher-order models.

Cannot independently mean: vocal effort, fatigue, stress, cognitive load, or illness.

EV_PHO_002 – HARMONICITY_SHIFT

Qualified HNR or related harmonicity change.

Must be interpreted in the context of recording conditions and vocal level.

EV_PHO_003 – PERTURBATION_SHIFT

Qualified change in jitter and/or shimmer.

STRICT GATE

This marker is eligible only when:

- the underlying segment meets periodicity requirements;
- F0 tracking is reliable;
- intensity-related confounding is accounted for;
- the extractor is validated for the relevant speech material.

Jitter and shimmer SHALL NOT be mandatory evidence for spontaneous connected speech.

EV_PHO_004 – PHONATORY_STABILITY

Composite Evidence representing relative stability of qualified voice-quality measures.

Candidate inputs:

- CPP/CPPS;
- HNR;
- eligible perturbation measures;
- local temporal consistency.

Requires multiple compatible indicators where possible.

10. Spectral Evidence Family

EV_SPE_001 – SPECTRAL_BALANCE_SHIFT

Change in compatible spectral-balance descriptors.

Candidates include:

- alpha ratio;
- Hammarberg index;
- spectral slopes;
- selected harmonic-difference measures.

May support: calibrated Activation, Processing Demand, and phonatory-change models.

It SHALL NOT independently create semantic meaning.

EV_SPE_002 – SPECTRAL_DYNAMICS_SHIFT

Change in spectral flux or other registered spectral-dynamics measures.

May support: modulation breadth and contextual reconfiguration.

EV_SPE_003 – ARTICULATORY_PATTERN_SHIFT

Change in formant-based measurements after compatible phonetic/language normalization.

v0.1 status

RESEARCH_ONLY

Unconstrained F1/F2/F3 changes are heavily dependent on the words and vowels spoken and SHALL NOT enter production semantic inference without an approved normalization method.

11. Dynamic Evidence Family

EV_DYN_001 – WITHIN_PROMPT_DRIFT

Composite marker indicating a consistent directional trajectory within one prompt.

Eligible components may include:

- F0 drift;
- loudness drift;
- CPP drift;
- rate drift;
- pause-load trajectory.

A drift marker requires sufficient time coverage.

EV_DYN_002 – WITHIN_PROMPT_VOLATILITY

Degree of short-timescale variation across qualified acoustic families.

This describes acoustic volatility.

It is not emotional volatility.

EV_DYN_003 – CONTEXT_RECONFIGURATION

Evidence that the acoustic organization changed when the prompt context changed.

Primary transitions:

$$P1 \rightarrow P2$$

$$P1 \rightarrow P3$$

$$P2 \rightarrow P3$$

Requires changes in at least two independent acoustic families before it may become a strong composite marker.

EV_DYN_004 – CONTEXT_CONSISTENCY

Evidence that the qualified acoustic organization remained comparatively similar across changing prompt contexts.

Context consistency is not automatically positive.

It may represent stable organization, limited reconfiguration, or simply a speaker whose vocal behavior is comparatively invariant.

Downstream meaning determines which interpretation, if any, is permitted.

EV_DYN_005 – MODULATION_BREADTH

Breadth of variation expressed across multiple acoustic domains.

Candidate domains:

- prosody;
- energy;
- timing;
- phonatory quality;
- spectral dynamics.

This is a descriptive multimodal-acoustic marker.

It does not mean creativity or emotional openness.

EV_DYN_006 – CONTEXTUAL_DEPLOYMENT_SHIFT

Change in how much of the speaker's observable acoustic range is deployed across prompts.

This marker is particularly relevant to Expressive Interface.

It requires evidence of an available range before compression can be interpreted as relative deployment.

EV_DYN_007 – MULTI_FAMILY_COHERENCE

Evidence that independent acoustic families move in a mutually compatible pattern under an active rule.

This is rule-specific.

There is no universal acoustic direction that constitutes coherence.

EV_DYN_008 – MULTI_FAMILY_DISCORDANCE

Evidence that candidate supporting families materially disagree.

High discordance reduces semantic confidence.

It may force:

UNRESOLVED

even when individual effects are large.

EV_DYN_009 – SHORT_INTERVAL_MAINTENANCE

Evidence that speech-production organization remains comparatively maintained over the observed prompt interval.

Candidate components:

- continuity;
- low disruptive drift;
- phonatory stability;
- rate maintenance;
- qualified energy/prosodic trajectory.

Scope is strictly limited to the observed interval.

12. Evidence independence

Multiple measurements derived from the same acoustic phenomenon do not automatically count as independent evidence.

For example:

```
F0 mean  
F0 median  
F0 percentile
```

are not three independent Evidence families.

Likewise:

```
pause load  
pause duration  
pause density
```

provide richer timing information but remain within the Timing family.

Abstract semantic parameters SHALL require evidence across independent families where their registry specifies multiple-family support.

13. Evidence-family codes

Canonical families:

```
PRO = Prosody  
ENG = Energy
```

```
TIM = Timing
PH0 = Phonation / voice quality
SPE = Spectral
DYN = Temporal / cross-context dynamics
```

Quality metadata is not itself semantic Evidence.

14. Prompt scopes

An Evidence Marker SHALL declare one or more scopes:

```
P1
P2
P3
P2_VS_P1
P3_VS_P1
P3_VS_P2
WITHIN_PROMPT
ALL_PROMPTS
PERSONAL_LONGITUDINAL
```

A rule cannot consume a marker outside its authorized prompt scope.

15. Confound handling

Confound flags include, at minimum:

```
MICROPHONE_CHANGE
DEVICE_CHANGE
AGC_SUSPECTED
BACKGROUND_NOISE
CLIPPING
LOW_VOICED_CONTENT
ILLNESS_REPORTED
VOICE_CONDITION_REPORTED
LANGUAGE_INCOMPATIBILITY
EXTRACTOR_INCOMPATIBILITY
PROMPT_NONCOMPLIANCE
INTERRUPTION
UNSTABLE_REFERENCE
```

A confound may:

- reduce confidence;
- suppress a component;
- suppress a marker;
- force a downstream dimension unresolved.

It may never be silently ignored.

16. Evidence publication rule

No Evidence Marker may be shown as a user-facing conclusion merely because its numeric value is extreme.

The Evidence Ledger distinguishes:

MEASURED
QUALIFIED
INTERPRETATION_ELIGIBLE
USER_EXPLAINABLE

These are separate states.

17. Evidence-to-Dimension boundary

The following is prohibited:

single measurement $\rightarrow$ semantic conclusion.

Preferred structure:

multiple qualified measurements $\rightarrow$ descriptive Evidence $\rightarrow$ independent Evidence families $\rightarrow$ bounded functional

Higher-level dimensions may specify minimum independent Evidence-family coverage.

18. Scientific implementation notes

The standardized acoustic foundation is informed by the GeMAPS/eGeMAPS approach: use a compact, interpretable acoustic vocabulary rather than assuming that a larger feature count automatically produces better scientific meaning [S1].

Controlled cognitive-load studies have reported changes across acoustic measures such as CPP, spectral energy relationships, F0, and sound-pressure level, but those findings support multivariate evidence—not a universal direct acoustic-to-cognitive-state mapping [S2–S3].

Natural autobiographical emotional speech shows considerable inter-speaker variability and relatively weak/simple valence effects, supporting SoulScope's decision to treat its three prompts as contexts rather than emotion ground-truth labels [S4].

Perturbation and harmonicity measurements require strict qualification because vocal intensity can materially influence jitter, shimmer, and HNR [S5].

19. References

[S1] Eyben, F., et al. *The Geneva Minimalistic Acoustic Parameter Set (GeMAPS) for Voice Research and Affective Computing*. IEEE Transactions on Affective Computing.

[S2] MacPherson, M. K., et al. *Acoustic Measures of Voice and Physiologic Measures of Autonomic Arousal During Speech as a Function of Cognitive Load*. Journal of Voice.

[S3] Abur, D., et al. *Acoustic Measures of Voice and Physiologic Measures of Autonomic Arousal During Speech as a Function of Cognitive Load in Older Adults*. Journal of Voice.

[S4] Delvaux, V., et al. *An Acoustic Study of Natural Emotional Speech Productions*. Interspeech 2022.

[S5] Brockmann-Bauser, M., et al. *Acoustic Perturbation Measures Improve with Increasing Voice Sound Pressure Level in Individuals With and Without Voice Disorders*. Journal of Voice.

20. Final Evidence rule

Evidence describes the signal before SoulScope attempts to describe the person.